Project Planning Document
Coupon Redemption Prediction & Promotion Targeting

Nhom 11 — FPT University

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Abstract. This document presents the overall project plan for *Project 11: Coupon Redemption Prediction & Promotion Targeting*. The project addresses the business problem of wasteful mass-coupon disbursement by building a machine-learning pipeline — anchored on RFM segmentation and XGBoost-based uplift modelling — that predicts individual coupon redemption probability. The plan covers six execution steps, a ten-week roadmap, visualisation strategy, API/dashboard integration, research-based learning requirements, and an AI audit framework. This document serves as the authoritative planning reference; detailed technical deliverables are produced in subsequent phase reports.

Keywords: coupon redemption, uplift modelling, RFM segmentation, XGBoost, CausalML, promotion targeting, data science project planning.

0 Contents

1 Problem Understanding & Data Collection

1.1 Business Problem

Mass coupon campaigns incur significant budget waste because coupons are distributed indiscriminately to customers who would have purchased anyway. The objective of this project is to **predict which customers will redeem coupons**, enabling highly precise, cost-efficient targeting.

1.2 Data Sources

- **Primary dataset:** Online Retail II (Kaggle) — columns include `InvoiceNo`, `CustomerID`, `Quantity`, `UnitPrice`, `InvoiceDate`, among others.
- **Simulated coupon table:** Derived from the primary dataset; columns include `DiscountValue`, `CouponType`, and `IsRedeemed`.

1.3 Target & Feature Variables

- **Target variable:** `IsRedeemed` (binary: 1 = redeemed, 0 = not redeemed).
- **Feature variables:** RFM metrics (Recency, Frequency, Monetary) and coupon-level attributes.

1.4 Analytical SQL Queries

Three SQL queries are required at this stage:

1. Redemption rate segmented by coupon type.
2. Average discount value across the dataset.
3. Count of repeat coupon users.

1.5 Self-Study Topics

RFM segmentation, promotional mechanics, Kaggle dataset workflows, dimensional modelling.

1.6 Deliverables

- Project charter
- Data dictionary
- 3 SQL code snippets

1.7 AI Prompt Guidance

Ask AI to define the business problem, relevant KPIs (including ROI), and the precise target variable.

2 Data Cleaning & Exploratory Data Analysis

2.1 Data Cleaning

- Handle missing `CustomerID` records (deletion vs. imputation decision).
- Cap `Quantity` and `UnitPrice` at the 99th percentile to address outliers.

2.2 Feature Engineering

Compute Recency, Frequency, and Monetary (RFM) metrics from purchase histories to construct a `Master_CustomerRFM` table.

2.3 Data Splitting

Chronologically split the dataset:

$$\text{Training set} = 80\% \text{ (earlier period)}, \quad \text{Test set} = 20\% \text{ (later period)}.$$

No random shuffling is permitted.

2.4 Exploratory Data Analysis

Run separate descriptive statistics for *coupon redeemers* vs. *non-redeemers* to surface baseline behavioural differences.

2.5 Self-Study Topics

Missing data mechanisms, outlier detection, chronological train/test splitting, SMOTE for class imbalance.

2.6 Deliverables

- Cleaned CSV data file
- RFM feature files
- EDA report
- Time-aligned train/test split files

2.7 AI Prompt Guidance

Inquire about the optimal strategy for rows with missing `CustomerID` — deletion versus imputation.

3 Advanced Data Visualisation

3.1 Required Charts

1. **Uplift Histogram:** Distribution of predicted uplift scores to assess model classification quality.

2. **RFM Metrics Comparison:** Side-by-side comparison of R, F, and M values between redeemers and non-redeemers.

3. **ROC Uplift Curve & QINI Chart:** Evaluate discriminative power; trace cumulative gains and profits from targeting the right audience.

4. **Cost-Benefit Analysis Table:** The most critical visual — calculates ROI by laying out explicit costs, revenues, and net profit across customer groups.

3.2 Self-Study Topics

Uplift score distributions, QINI curves, Gain charts, profit-driven visualisations.

3.3 Deliverables

- 4 chart files (PNG format)

- Cost-benefit analysis table
- Optional: Power BI dashboard

3.4 AI Prompt Guidance

Request code to generate a QINI curve and request an explanation of its direct business implications.

4 Service Integration & API Development

4.1 Power BI Data Pipeline

Set up a Python script to push `predictions.csv` directly into Power BI for live dashboard refresh.

4.2 Dashboard Design

A 3-page Power BI layout:

1. ROI Tracking
2. High-Priority Target List (top 20% of customers by uplift score)
3. RFM Segment Analysis

4.3 API Development (Optional)

Build a Flask REST API endpoint that accepts a `CustomerID` and returns:

- Real-time uplift score
- Customer segment label

4.4 Self-Study Topics

Python connectors in Power BI, DAX for custom profit calculations, Slicers, Flask API fundamentals.

4.5 Deliverables

- 3-page `.pbix` file
- `predictions.csv` file
- Flask backend code (optional)
- 2-minute demo video

4.6 AI Prompt Guidance

Request Python code to export prediction files, step-by-step instructions to load them into Power BI, and the DAX measures required to compute ROI.

5 Predictive Modelling

5.1 Baseline Model

Run Logistic Regression using RFM and coupon features to quantify how strongly discount value influences redemption probability.

5.2 Core Uplift Model

Deploy **XGBoost** via the `CausalML` library to produce an individual uplift score for every customer — defined as the incremental probability gain attributable to receiving a coupon:

$$\tau(x) = P(Y = 1 \mid T = 1, X = x) - P(Y = 1 \mid T = 0, X = x)$$

5.3 Evaluation & Optimisation

- Evaluation metrics: QINI curve, AUUC, Precision@K.
- Hyperparameter tuning via time-series cross-validation.

5.4 Model Comparison

Compare three approaches using AUUC and Precision@K:

1. Baseline Logistic Regression
2. XGBoost Uplift Model
3. Random targeting (baseline reference)

5.5 Self-Study Topics

Uplift modelling framework (CATE), XGBoost algorithm internals, `CausalML` library, meta-learner variants (S-learner, T-learner, X-learner).

5.6 Deliverables

- Model training and evaluation scripts (`.py`)
- Feature importance charts
- Saved model files (`.pkl`)

5.7 AI Prompt Guidance

Request deployment details for an XGBoost uplift classifier using CausalML and an explanation of the uplift loss function.

6 AI Application Development

6.1 Interactive Targeting Tool

Integrate a *what-if* slider into the dashboard so stakeholders can dynamically adjust coupon discount values and observe projected profit fluctuations in real time.

6.2 Automated AI Recommendations

Build a rules layer that automatically labels each customer as **Target** or **Do Not Target** based on their model-assigned uplift score.

6.3 Executive Summary View

Construct an executive-facing dashboard page displaying:

- Total budget utilisation
- ROI gains over traditional mass-mailing strategies

6.4 Self-Study Topics

Power BI what-if parameters, Streamlit deployment, exporting model objects via `pickle`.

6.5 Deliverables

- Finalised dashboard (Power BI or Streamlit)
- Clean 1-page user guide
- Presentation slide deck

6.6 AI Prompt Guidance

Design a what-if slider parameter in Power BI to monitor profit fluctuations dynamically.

7 Research-Based Learning (RBL) Citation Guide

The final Springer-formatted LaTeX report must formally cite **at least 3 peer-reviewed papers**. The following search strategy is recommended:

1. **Uplift Modelling Foundation.** Search keywords: *"uplift modeling" AND "coupon"*. Priority paper: *Uplift Modeling for Coupon Redemption*.

2. **CausalML & XGBoost.** Search keywords: *"causalml" AND "marketing campaign"*. Essential reference: Athey & Imbens (2016).

3. **RFM & Coupon Redemption.** Focus on publications from 2020–2025 using keywords: *"RFM segmentation" AND "coupon redemption"*.

8 AI Audit Log

Maintain an Excel workbook with **four dedicated sheets** to track and audit all AI-assisted work: *Core Prompts*, *AI Responses*, *Hallucination Detections*, and *Human Delta*.

Pay particular attention to:

- AI inventing non-existent CausalML API functions.
- Faulty or incorrectly labelled QINI curves.
- Synthetic data that lacks core retail business logic.

Table 1. AI Audit Log — illustrative schema.

Field	Illustrative Example
Date	2026-05-13
Step	Step 5 — Predictive Modelling
Prompt	*"Deploy XGBoost uplift classifier. . . "*
Human Delta	AI suggested default hyperparameters; manually added time-series cross-validation due to strong seasonal patterns in the retail data.

Field	Illustrative Example
Hallucination?	Yes/No — If Yes, explicitly describe how it was caught and corrected.

9 10-Week Execution Roadmap

Table 2. Planned weekly schedule for Project 11.

Week	Phase	Core Focus & Activity	Deliverables
1	Step 1	Problem analysis, data download, SQL querying, data dictionary creation.	Project charter + 3 SQL views.
2	Step 2	Data cleaning, handling missing/outlier values, engineering initial RFM features.	Cleaned CSV + EDA report.
3	Steps 2–3	Finalise EDA; begin plotting uplift distributions and RFM comparison charts.	EDA documentation + 2 charts.
4	Step 3	Complete all 4 charts, build cost-benefit table, read Research Paper #1.	Full chart set + RBL notes.
5	Step 5	Run Baseline Logistic Regression; train core Uplift model with CausalML.	`model_training.py`
6	Step 5	Evaluate models (QINI, Precision@K) and fine-tune hyperparameters.	Evaluation report + `.pkl` files.
7	Step 4 + RBL	Integrate Power BI, set up data pipeline, read Research Papers #2 & #3.	`.pbix` draft + 3 citations.
8	Step 6	Finalise dashboard, integrate what-if slider, organise page layouts.	Completed `.pbix` file.
9	All	Write formal academic report in LaTeX (10–12 pages, Springer format).	Completed `.tex` + `.pdf`.
10	All	Finalise AI Audit Log entries; record project presentation video.	`.xlsx`, `.pptx`, `.mp4` files.

9.0 Risk Management Notes

9 References

[1] Author(s) TBD. *Uplift Modeling for Coupon Redemption*. To be added in final report (search: "uplift modeling AND coupon").

[2] Athey, S., & Imbens, G. (2016). *Recursive Partitioning for Heterogeneous Causal Effects*. Proceedings of the National Academy of Sciences, 113(27), 7353–7360.

[3] Author(s) TBD. *RFM Segmentation & Coupon Redemption* (2020–2025). To be added in final report (search: "RFM segmentation AND coupon redemption").